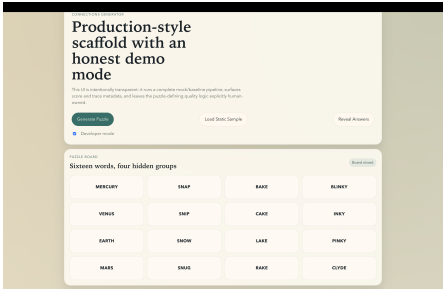
Connections Puzzle Generator

System Scaffold, Quality-Control Plan, and Implementation Handoff

A production-style system for generating, evaluating, and debugging NYT-Connections-style puzzles.



- End-to-end demo mode already runs.
- Quality-control infrastructure is scaffolded.
- Puzzle-quality heuristics remain intentionally human-owned.

Presenter: [Presenter Name] **Team:** [Team or Group] **Context:** [Course / Internal Handoff]

What This Project Is

Puzzle Format

- 16 words
- Partitioned into 4 groups of 4
- Each group has an internal connection

Our Task

- Generate puzzles
- Not just solve or classify them
- Produce puzzles that survive quality control

Why Generation Is Hard

Good local groups are not enough; the whole puzzle must avoid multiple plausible solutions.

Current Product Status

Frontend Surface

React + TypeScript + Vite
Puzzle board, reveal panel, score panel, debug view

Backend + Pipeline

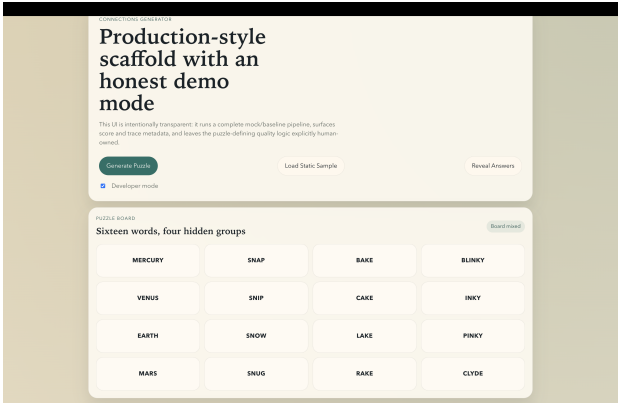
FastAPI, typed schemas, services, demo generation, trace metadata

Evaluation + Debug

Batch evaluation, accepted/rejected persistence, top-k browsing

This repo is already a real product scaffold, not just disconnected code fragments.

Demo and Product Surfaces



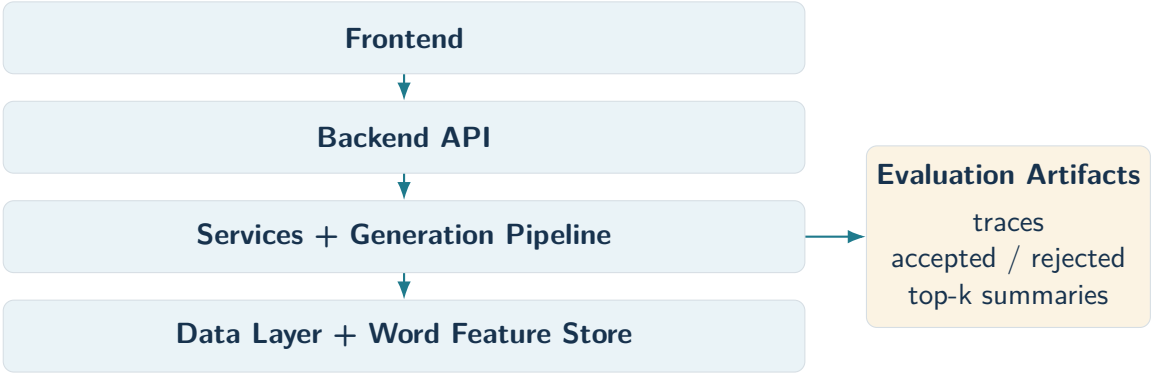
What the UI Already Exposes

- Generate Puzzle
- Load Static Sample
- Reveal Answers
- Developer Mode
- Top-K / batch debug summary

Demo Message

We can show both the product-facing flow and the engineering-facing debug flow in one interface.

Big-Picture Architecture



Frontend on top, pipeline in the middle,
reusable data and evaluation artifacts underneath.

Main Generation Pipeline

1. **Build a candidate**

Word Feature DB → Group Generators → Puzzle Composer

2. **Judge the candidate**

Solver → Ambiguity → Verifier → Scorer → UI / Batch Eval

The project is not just a generator; it is a generator plus a quality-control pipeline.

Four Generator Families

Semantic

Main stable generator.
Best first target for strong, reliable groups.

Lexical

Pattern and template diversity.
Helps avoid every puzzle feeling semantically uniform.

Theme

NYT-like thematic richness.
Important for memorable, human-feeling puzzles.

Phonetic / Wordplay

High-upside category for cleverness.
Also the hardest and most later-stage generator.

Why Quality Control Matters

What Generation Can Get Right

- Each group looks internally coherent
- The categories feel individually plausible
- The final 16 words look interesting

What Can Still Go Wrong

- Words leak across groups
- Multiple plausible partitions exist
- The puzzle is technically valid but stylistically weak

Generating categories is only half the problem; the real challenge is controlling ambiguity at the full-puzzle level.

Ambiguity Evaluator, Verifier, and Scorer

Ambiguity Evaluator

Goal: inspect structural risk.

Output: leakage evidence, alternative grouping candidates, risk level.

Verifier

Goal: act as the hard gate.

Output: accept, reject, or borderline with explicit reasons.

Scorer

Goal: rank what survives.

Output: coherence, ambiguity penalties, style metadata, overall score.

These modules do different jobs: evidence, gating, and ranking should not be collapsed into one heuristic blob.

Second-Stage Scaffold Already Completed

What the repo already supports today	
Solver ensemble	Multiple solver adapters, per-solver votes, agreement/disagreement summaries
Ambiguity scaffold	Structured leakage evidence, alternative grouping records, risk reports
Style-analysis scaffold	Placeholder style signals, archetype labels, transparent non-final reporting
Batch evaluation	Offline runs, accepted/rejected persistence, summary metrics, saved top-k outputs
Top-k debug	Developer-facing browsing for best accepted puzzles and comparison metadata

Important boundary

The infrastructure exists. The true puzzle-quality heuristics do not.

Implemented Scaffold vs Human-Owned Core Logic

Already Implemented

- FastAPI backend and React frontend
- Demo generation pipeline
- Typed schemas and trace metadata
- Solver registry and ensemble scaffold
- Ambiguity, style, batch-eval, and top-k scaffolds

Still Intentionally Human-Owned

- High-quality feature engineering
- Real generator quality for all four families
- Leakage reasoning and alternative partition search
- Final verifier policy and score calibration
- Real NYT-likeness and style judgment

Implementation Roadmap

Recommended order of execution		
Phase 1	**Representation + generation core**	Semantic feature DB, semantic generator, and puzzle composer
Phase 2	**Quality gate**	Ambiguity evaluator, verifier, and scorer
Phase 3	**Diversity**	Lexical generator and theme generator
Phase 4	**Tuning loop**	Batch evaluation, analysis, and acceptance tuning
Phase 5	**A+ layer**	Phonetic generator, style calibration, final polish

Build the stable semantic core first; add richer diversity and polish only after the quality gate exists.

MVP vs Strong Version vs A+ Enhancements

MVP

- Semantic feature DB
- Semantic generator
- Composer
- Ambiguity + verifier + scorer
- Batch eval + web UI

Strong Version

- Better semantic coverage
- Better composer constraints
- More reliable ranking
- Stronger tuning loop

A+ Enhancements

- Theme depth
- Phonetic / wordplay groups
- Historical style calibration
- Better solver ensemble and UI polish

Team Action Slide: What Must Be Owned Next

Core build work

- Feature extraction: define semantic, lexical, phonetic, and theme-oriented signals
- Generator quality: make semantic strong first, then add lexical and theme diversity
- Composer: enforce deduplication, compatibility, and puzzle-level rules
- Ambiguity: turn scaffolded evidence into real leakage reasoning and regrouping search

Quality gate and tuning

- Verifier: define the accept / reject / borderline policy clearly
- Scorer: rank accepted puzzles with calibrated, explainable signals
- Evaluation and calibration: use batch runs, top-k review, and traces to improve quality over time

Closing Takeaway

Why this project is compelling

- It already has a credible product surface and engineering scaffold.
- It is honest about what is solved and what is not.
- The remaining work is meaningful research-engineering, not basic plumbing.

The scaffold is ready; the next step is to turn human-owned judgment into measurable puzzle quality.